

Introduction to Data Science Group Project Task 2

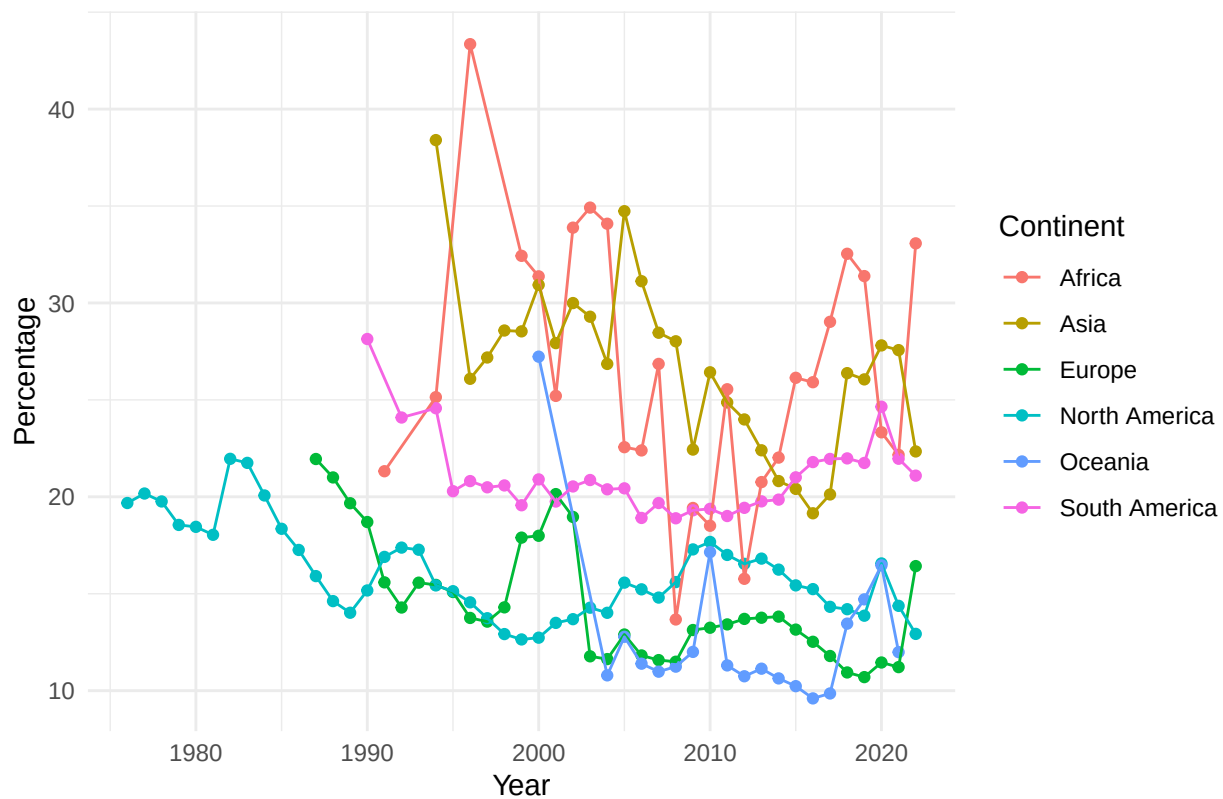
K9

2025-12-08

In this section, we will discuss the extent each continent working toward the goal of reducing the proportion of youth not in employment, education or training (hereafter referred to as NEET).

We would first draw a graph to demonstrate the change of NEET for each continent with the year using the line graph.

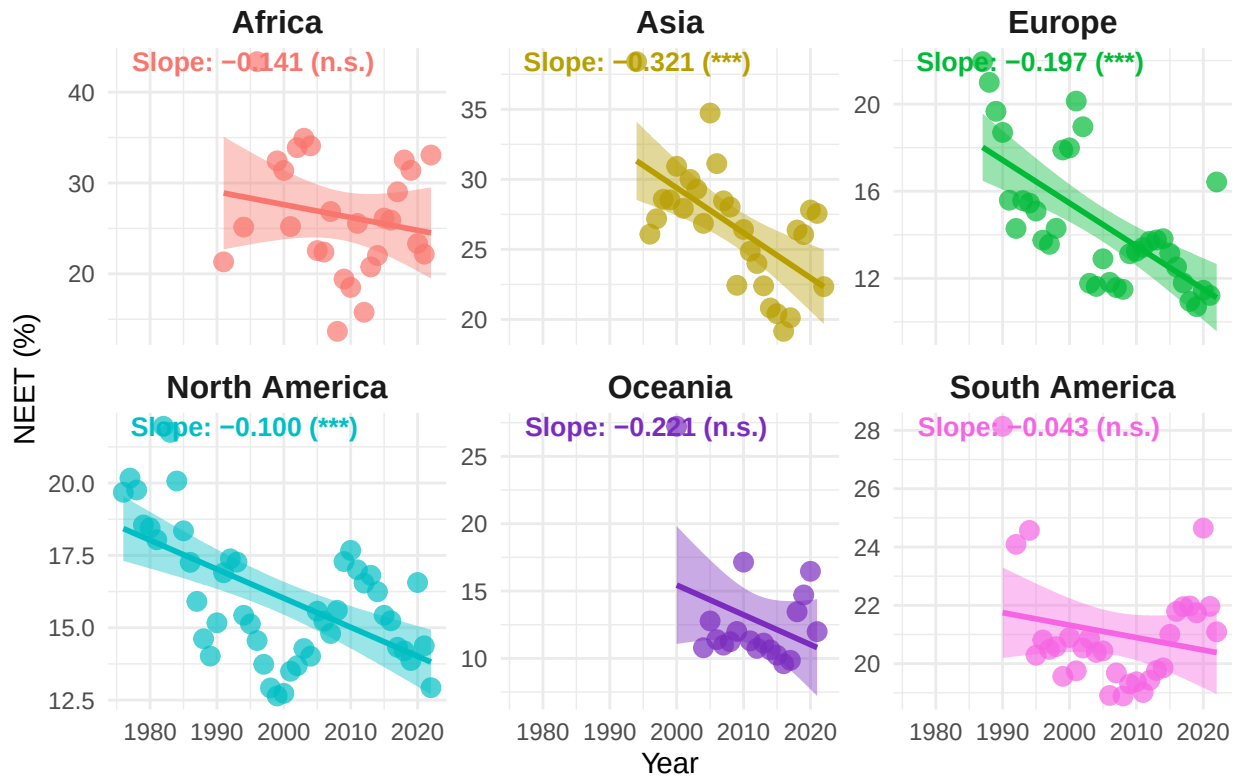
Figure 1



We then would like to see the regression line for the NEET.

```
## `geom_smooth()` using formula = 'y ~ x'
```

Figure 2



Then we would like to see how education and employment each contribute to the EET, one would create a table to illustrate how they correlated

Table 1: Continental Correlation

Continent	Corr_SE	P_Value_SE	Corr_Unemp	P_Value_Unemp
Africa	0.2232	0.0156	0.4211	0.0000
Asia	-0.3821	0.0000	0.5954	0.0000
Europe	-0.3373	0.0000	0.6834	0.0000
North America	-0.7692	0.0000	-0.0507	0.4635
Oceania	-0.8927	0.0000	0.2500	0.0902
South America	-0.4351	0.0000	0.5748	0.0000

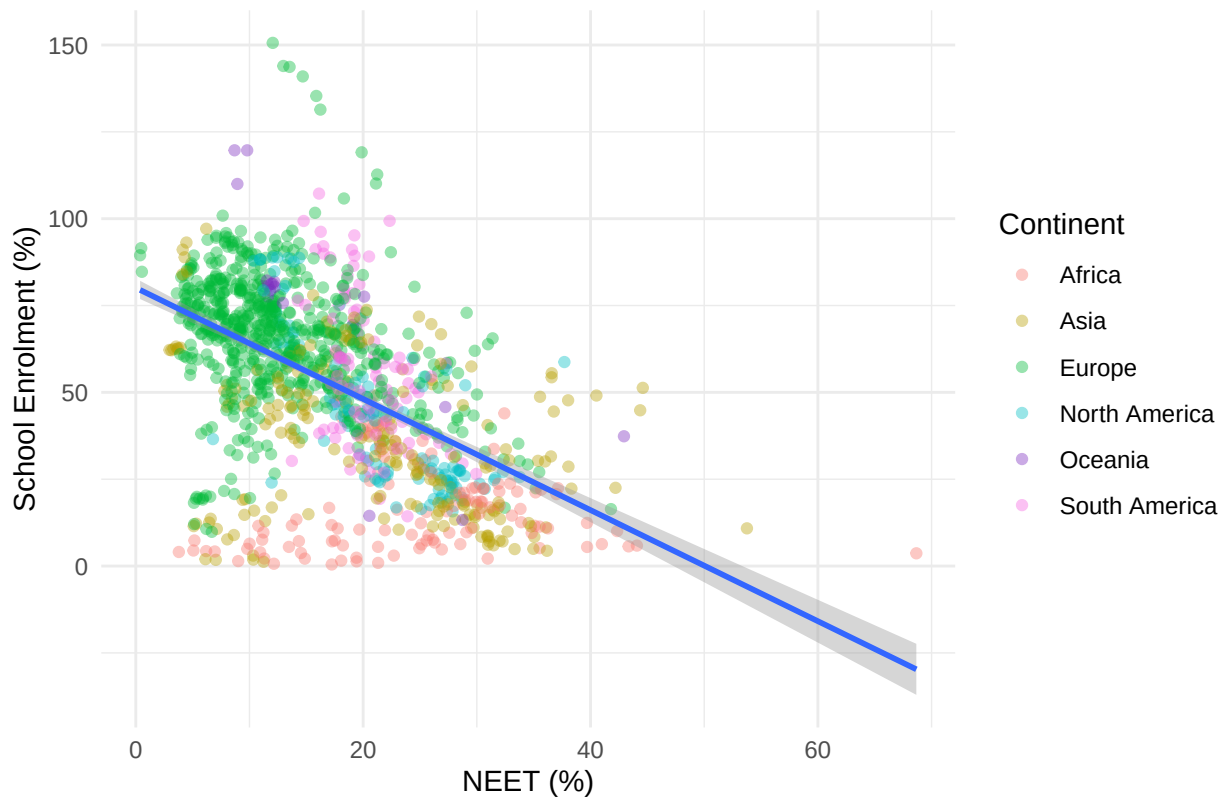
If we do not group the data by continent rather seeing them as a whole, then we get the following table

Table 2: Global Correlation

variable	correlation	p_value
Unemployment	0.3161	0
SE	-0.5574	0

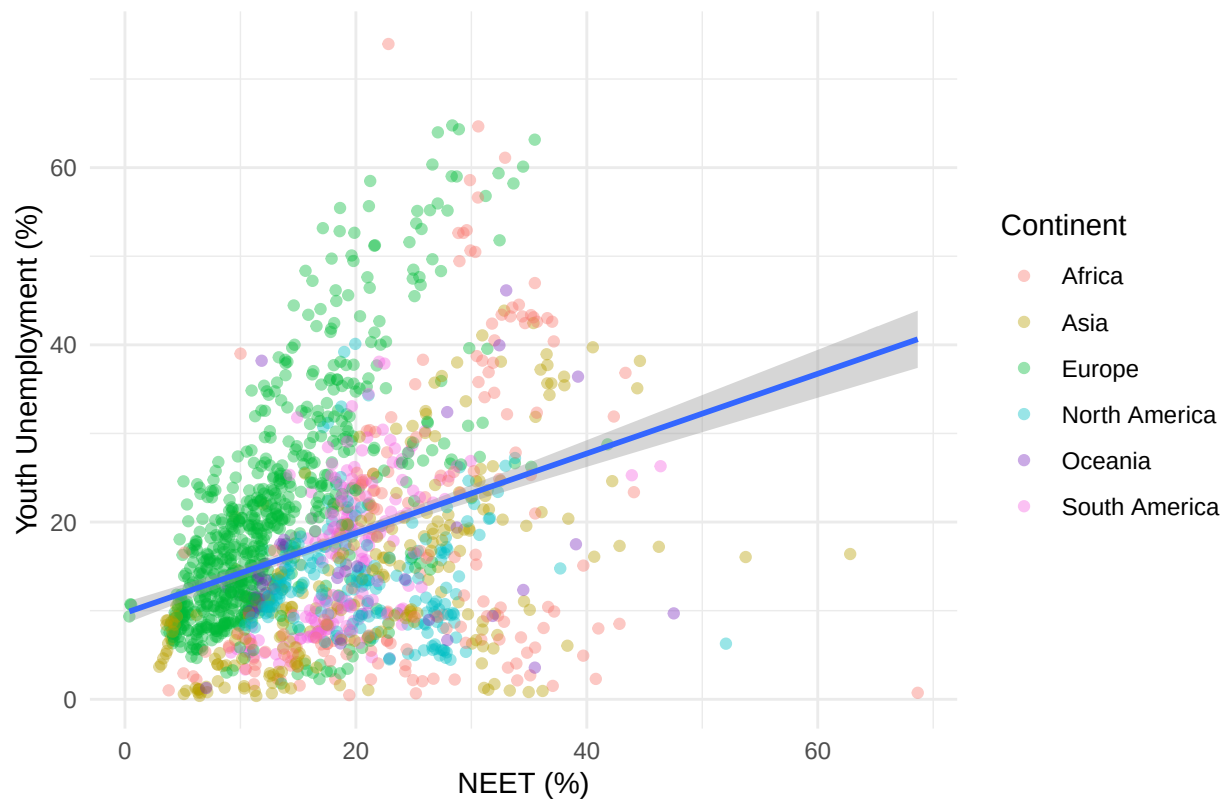
```
## `geom_smooth()` using formula = 'y ~ x'
```

School Enrolment (Tertiary) and NEET Percentage Comparison



```
## `geom_smooth()` using formula = 'y ~ x'
```

Youth Unemployment and NEET Percentage Comparison



The below code chunks would be focus on the correlation between NEET and GDP per head.

Global Correlation

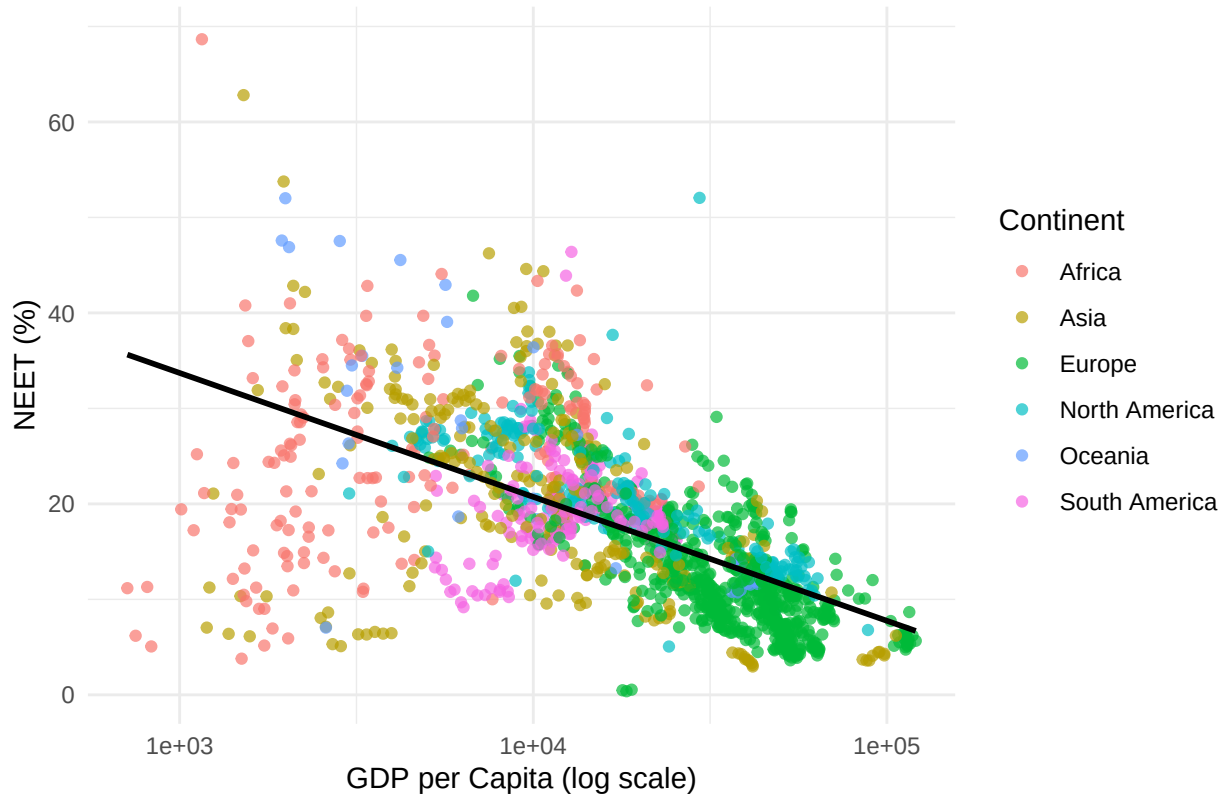
```
##
## Pearson's product-moment correlation
##
## data:  gdp_neet$Gdp_per_capita and gdp_neet$Percentage_NEET
## t = -32.692, df = 1550, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.6673823 -0.6084206
## sample estimates:
##      cor
## -0.6388386

##
## Spearman's rank correlation rho
##
## data:  gdp_neet$Gdp_per_capita and gdp_neet$Percentage_NEET
## S = 1058310214, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## -0.6985928
```

Global Plot

```
## `geom_smooth()` using formula = 'y ~ x'
```

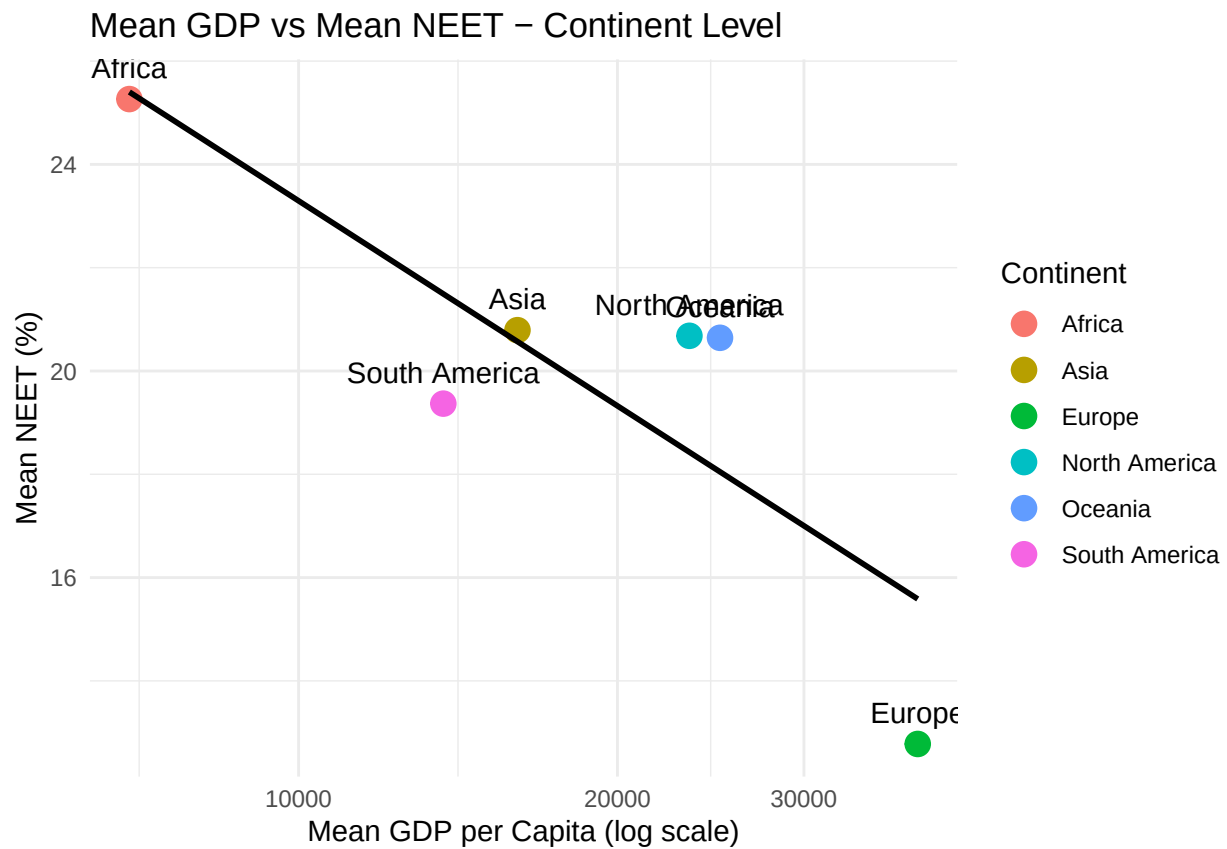
GDP per Capita vs NEET (%) – Global



Continent Means + Correlation

Continent Plot

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
##           Level  Pearson_r Spearman_r      p_value      N
## 1      Global (Countries) -0.6388386 -0.6985928 9.660679e-179 1552
## 2 Continent Means (6 points) -0.8816186 -0.6571429 2.019173e-02   6
## Significant
## 1      TRUE
## 2      TRUE
```